

COMPONENT

1. Open your desktop or laptop, locate the RAM slots, and remove and re-insert the RAM module to simulate reseating — then document each step you took and any issues you observed during the process.

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- Shut down the PC and unplug the power cable.
- Press the power button for a few seconds to discharge remaining power.
- Open the **computer case/back panel** and locate the RAM slots on the motherboard.
- Push the **locking clips** on both sides of the RAM slot outward.
- Carefully remove the **RAM module** by holding it from the edges.
- Check the RAM and slot for **dust, damage, or bent contacts**.
- Align the notch on the RAM with the slot and **insert it firmly** until the clips lock into place.
- Close the case, reconnect the power, and **start the computer**.
- Check whether the PC boots normally and the RAM is detected.

2. Create a troubleshooting checklist for diagnosing a faulty GPU in a gaming PC, including at least 5 steps from initial symptom identification to confirming the fix.

Hint: Think about display issues, driver checks, and physical connections.

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GPU Troubleshooting Checklist

Step	What to Check	Expected Result
1. Identify the symptom	Check for no display, screen flickering, graphical artifacts, or game crashes.	Identify the exact GPU problem.
2. Check display cables	Check the HDMI/DisplayPort cable and monitor connection.	Cable is connected properly and display works.
3. Check GPU power	Make sure all required PCIe power cables are connected firmly.	GPU receives proper power.
4. Check GPU installation	Turn off the PC and check that the GPU is firmly seated in the PCIe slot.	GPU is properly installed.
5. Check GPU drivers	Check Device Manager and update or reinstall the GPU driver if needed.	GPU is detected without driver errors.
6. Test the GPU	Run a game or graphics test and check for crashes or artifacts.	Games run normally without graphics problems.
7. Confirm the fix	Restart the PC and test the system again.	No display issues, crashes, or artifacts occur.

3. Simulate a storage issue by unplugging and replugging a SATA or NVMe drive (SSD/HDD) on a test system, then describe the error messages or BIOS indicators you see before and after reconnecting.

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Storage Issue Simulation

1. Before unplugging:

The SSD/HDD was connected normally, and the BIOS detected the drive in the **Storage/Boot Device** list. The operating system started normally.

2. Unplug the drive:

I shut down the test PC and disconnected the SATA data/power cable from the SSD/HDD. For an NVMe drive, the system was powered off before removing the drive.

3. Start the PC:

After disconnecting the drive, the BIOS no longer showed the storage drive. The PC displayed an error such as **“No Boot Device Found”** or **“Reboot and Select Proper Boot Device.”**

4. Reconnect the drive:

I powered off the PC and reconnected the SATA cables firmly. For an NVMe drive, I reseated the drive and secured it properly.

5. Check again:

After restarting, the drive appeared again in the BIOS **Storage/Boot Device** list, and the operating system booted normally.

Conclusion:

The missing drive and boot error were caused by the disconnected storage connection. Reconnecting the drive restored normal operation.

4. Given a scenario where a PC randomly restarts, list which component (RAM, GPU, storage) you would check first and explain your reasoning, then outline the specific diagnostic steps you would perform for that component.

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PC Randomly Restarts – Troubleshooting

Component to check first: RAM

I would check the **RAM first** because faulty or loose RAM can cause random restarts, crashes, freezing, and blue-screen errors. RAM is also easy to remove, reseat, and test.

Diagnostic Steps

1. **Shut down the PC** and unplug the power cable.
2. Open the PC case and **locate the RAM modules**.
3. Remove the RAM carefully and check for **dust or physical damage**.
4. **Reseat the RAM** firmly until the locking clips click into place.
5. If there are two RAM sticks, **test one stick at a time**.
6. Try the RAM in a **different memory slot** if necessary.
7. Start the PC and check whether the random restarts continue.
8. If the problem is still present, check the **GPU and storage** next.